

Module -2

Step Response of Parallel RLC Circuits

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SENSE

Step response of a Parallel RLC Circuit

The step response is obtained by the sudden application of a dc source.

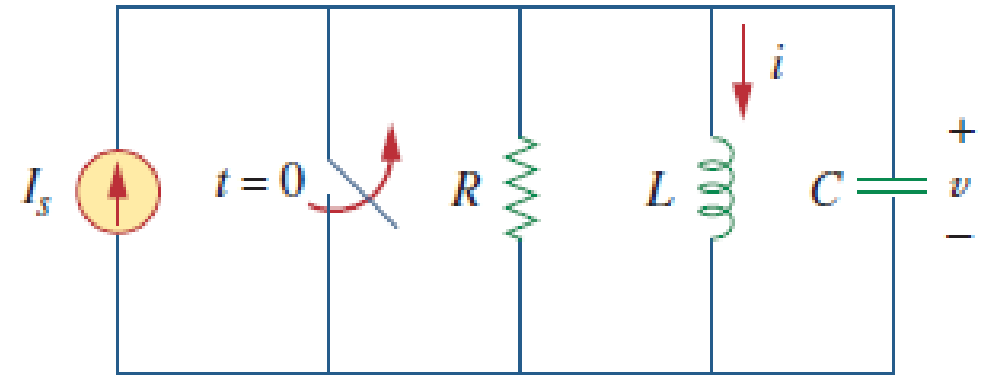
Applying KCL at the top node for $t > 0$,

$$\frac{v}{R} + i + C \frac{dv}{dt} = I_s \longrightarrow (1)$$

$$v = L \frac{di}{dt} \longrightarrow (2)$$

Substituting for v

$$\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC} \longrightarrow (3)$$



$$i(t) = i_t(t) + i_{ss}(t)$$

$$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

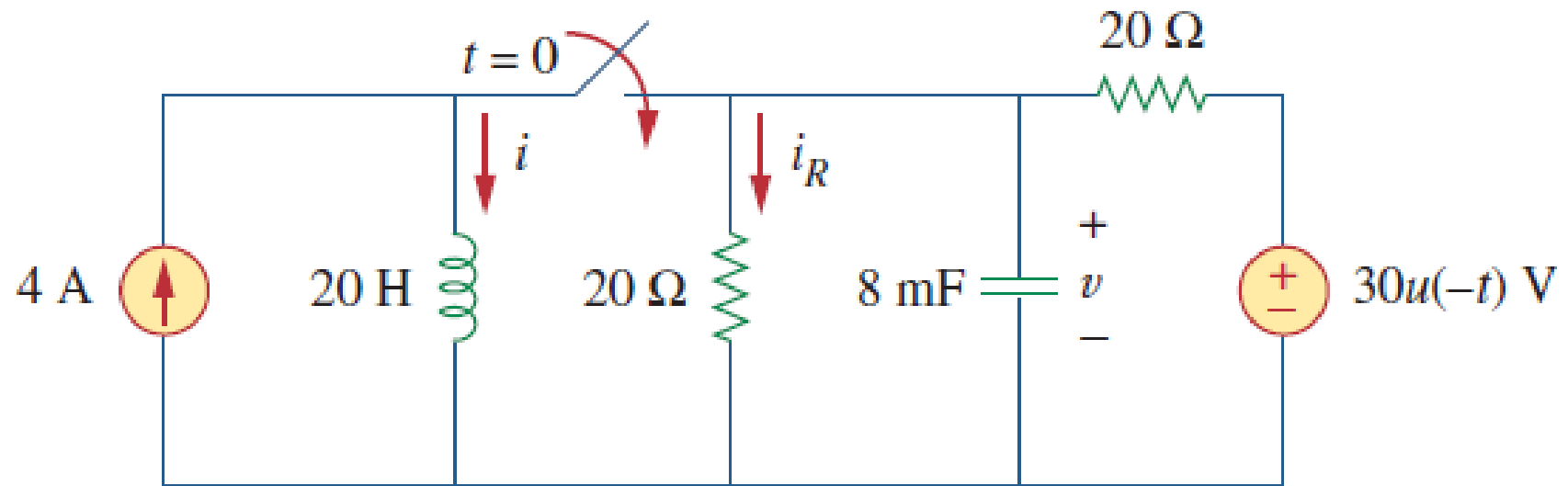
$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

The constants A_1 and A_2 can be determined from the initial conditions for i and di/dt .

Numerical - 1

Find $i(t)$ and $i_R(t)$ for $t > 0$.



Solution:

For $t < 0$, the switch is open, and the circuit is partitioned into two independent subcircuits. The 4-A current flows through the inductor, so that

$$i(0) = 4 \text{ A}$$

Since $30u(-t) = 30$ when $t < 0$ and 0 when $t > 0$,

The capacitor acts like an open circuit

$$v(0) = \frac{20}{20 + 20}(30) = 15 \text{ V}$$

For $t > 0$, the switch is closed, and we have a parallel RLC circuit with a current source. The voltage source is zero which means it acts like a short-circuit. The two $20\text{-}\Omega$ resistors are now in parallel. They are combined to give $R = 20 \parallel 20 = 10 \text{ }\Omega$. The characteristic roots are determined as follows:

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 10 \times 8 \times 10^{-3}} = 6.25$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20 \times 8 \times 10^{-3}}} = 2.5$$

$$\begin{aligned} s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -6.25 \pm \sqrt{39.0625 - 6.25} \\ &= -6.25 \pm 5.7282 \end{aligned}$$

$$s_1 = -11.978, \quad s_2 = -0.5218$$

Since $\alpha > \omega_0$, we have the overdamped case. Hence,

$$i(t) = I_s + A_1 e^{-11.978t} + A_2 e^{-0.5218t}$$

where $I_s = 4$ is the final value of $i(t)$. We now use the initial conditions to determine A_1 and A_2 . At $t = 0$,

$$i(0) = 4 = 4 + A_1 + A_2 \quad \Rightarrow \quad A_2 = -A_1$$

Taking the derivative of $i(t)$

$$\frac{di}{dt} = -11.978A_1e^{-11.978t} - 0.5218A_2e^{-0.5218t}$$

at $t = 0$,

$$\frac{di(0)}{dt} = -11.978A_1 - 0.5218A_2$$

$$L\frac{di(0)}{dt} = v(0) = 15 \quad \Rightarrow \quad \frac{di(0)}{dt} = \frac{15}{L} = \frac{15}{20} = 0.75$$

$$0.75 = (11.978 - 0.5218)A_2 \quad \Rightarrow \quad A_2 = 0.0655$$

Thus, $A_1 = -0.0655$ and $A_2 = 0.0655$. Inserting A_1 and A_2 in Eq.

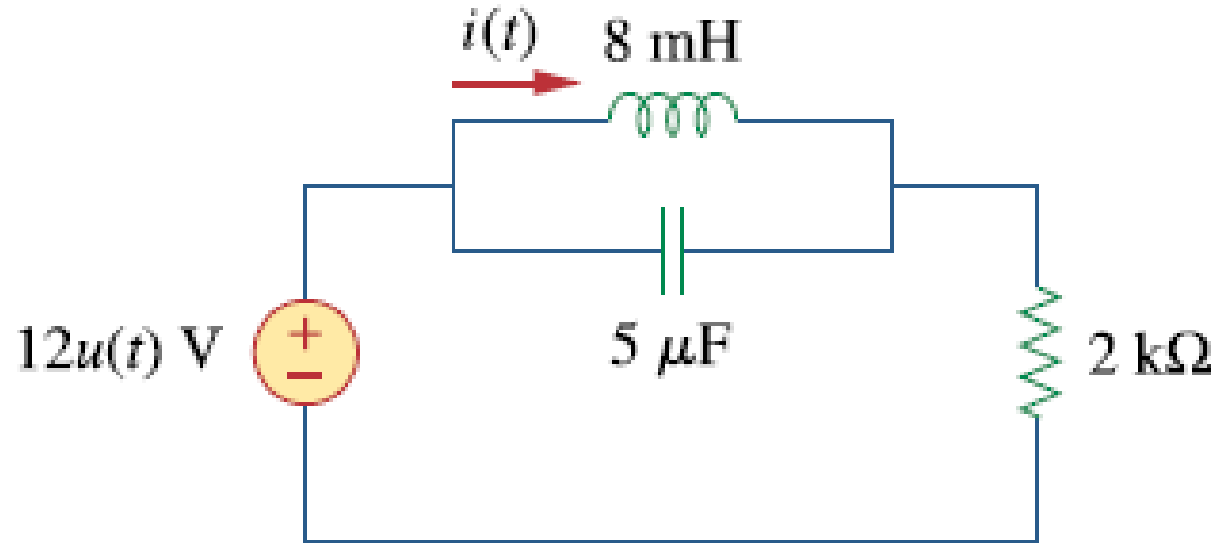
$$i(t) = 4 + 0.0655(e^{-0.5218t} - e^{-11.978t}) \text{ A}$$

From $i(t)$, we obtain $v(t) = L di/dt$ and

$$i_R(t) = \frac{v(t)}{20} = \frac{L}{20} \frac{di}{dt} = 0.785e^{-11.978t} - 0.0342e^{-0.5218t} \text{ A}$$

Numerical - 2

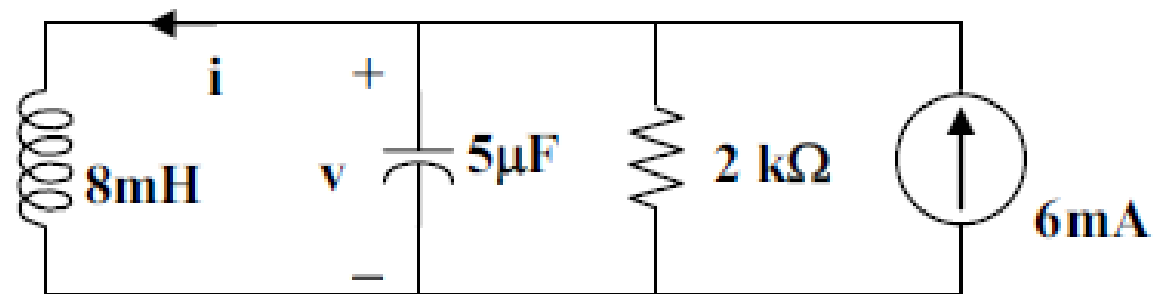
Find $i(t)$ for $t > 0$ in the circuit.



Solution

For $t = 0^-$, $u(t) = 0$, so that $v(0) = 0$ and $i(0) = 0$.

For $t > 0$, we have a parallel RLC circuit with a step input, as shown below.



$$\alpha = 1/(2RC) = (1)/(2 \times 2 \times 10^3 \times 5 \times 10^{-6}) = 50$$

$$\omega_o = 1/\sqrt{LC} = 1/\sqrt{8 \times 10^{-3} \times 5 \times 10^{-6}} = 5,000$$

Since $\alpha < \omega_o$, we have an underdamped response.

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2} \cong -50 \pm j5,000$$

$$i(t) = I_s + [(A\cos 5,000t + B\sin 5,000t)e^{-50t}], \quad I_s = 6\text{mA}$$

$$i(0) = 0 = 6 + A \quad \text{or} \quad A = -6\text{mA}$$

$$v(0) = 0 = Ldi(0)/dt$$

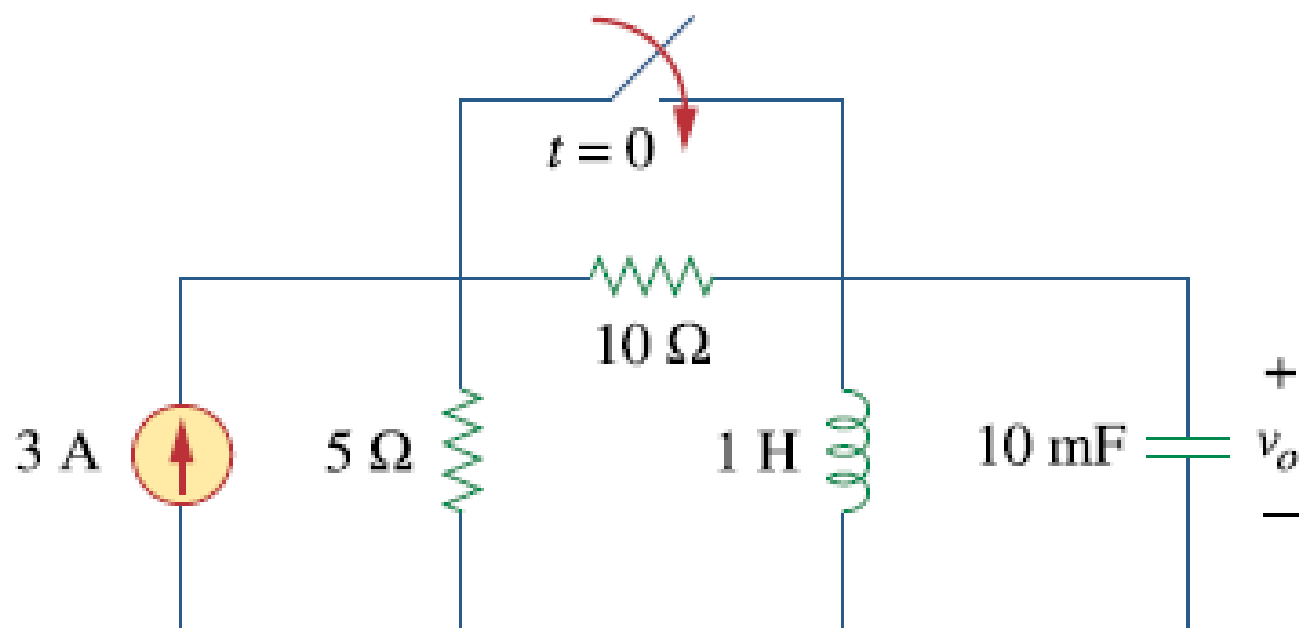
$$di/dt = [5,000(-A\sin 5,000t + B\cos 5,000t)e^{-50t}] + [-50(A\cos 5,000t + B\sin 5,000t)e^{-50t}]$$

$$di(0)/dt = 0 = 5,000B - 50A \quad \text{or} \quad B = 0.01(-6) = -0.06\text{mA}$$

$$i(t) = \underline{\{6 - [(6\cos 5,000t + 0.06\sin 5,000t)e^{-50t}]\} \text{ mA}}$$

Numerical - 3

Find the output voltage $v_o(t)$ in the circuit



Solution

At $t = 0^-$, we obtain, $i_L(0) = 3 \times 5 / (10 + 5) = 1 \text{ A}$

and $v_o(0) = 0$.

For $t > 0$, the 10-ohm resistor is short-circuited and we have a parallel RLC circuit with a step input.

$$\alpha = 1/(2RC) = (1)/(2 \times 5 \times 0.01) = 10$$

$$\omega_o = 1/\sqrt{LC} = 1/\sqrt{1 \times 0.01} = 10$$

Since $\alpha = \omega_o$, we have a critically damped response.

$$s_{1,2} = -10$$

$$i(t) = I_s + [(A + Bt)e^{-10t}], \quad I_s = 3$$

$$i(0) = 1 = 3 + A \quad \text{or} \quad A = -2$$

$$v_o = L di/dt = [Be^{-10t}] + [-10(A + Bt)e^{-10t}]$$

$$v_o(0) = 0 = B - 10A \quad \text{or} \quad B = -20$$

$$\text{Thus, } v_o(t) = \underline{(200te^{-10t}) \text{ V}}$$